Bridging the Gap: Statistical vs Machine Learning Methods

Performance, Cost, and Interpretability Trade-offs
in the Age of Transparent AI

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Abstract

This comprehensive study presents a systematic comparison between statistical and machine learning (ML) approaches across multiple problem domains. We evaluate 13 model-dataset combinations spanning four distinct problem types: regression, classification, time series analysis, and factorial analysis. Our analysis quantifies the fundamental trade-offs between three critical dimensions: performance (accuracy, R^2, RMSE), cost (training time, computational resources), and interpretability (explainability, statistical inference).

Our findings reveal that statistical models excel in interpretability (88.3/100 vs 37.4/100) and efficiency (20-100x faster training), while ML/DL models demonstrate superior performance on complex datasets (0.818 vs 0.547 average). The study connects these findings to the emerging field of Interpretable ML and Transparent AI, providing evidence-based recommendations for model selection. We demonstrate that no single approach dominates all categories, and the optimal choice depends on specific requirements including regulatory constraints, dataset characteristics, and computational resources.

Keywords: Machine Learning, Statistical Models, Interpretability, Transparent AI, Model Comparison, Performance Trade-offs

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1 Introduction

1.1 Background and Motivation

The landscape of predictive modeling has been fundamentally shaped by two distinct yet interconnected paradigms: traditional statistical methods and modern machine learning approaches. While statistical models have provided the foundation for scientific inference and hypothesis testing for over a century, machine learning methods have revolutionized predictive accuracy in complex, high-dimensional problems.

However, as artificial intelligence systems increasingly inform critical decisions in healthcare, finance, and public policy, the demand for **transparent and interpretable** models has reached unprecedented levels. This convergence of performance requirements and interpretability needs has created a fundamental tension: how can practitioners balance the superior predictive power of complex ML models with the transparency and trust required for high-stakes applications?

The emergence of **Interpretable ML** and **Transparent AI** research represents a concerted effort to bridge this gap. As noted in recent work on transparent AI [1], the challenge is not merely technical but encompasses organizational, regulatory, and human factors. This study contributes to this ongoing dialogue by providing empirical evidence on the trade-offs between statistical and ML approaches.

1.2 Problem Statement

The core research question driving this investigation is:

How do Machine Learning and Statistical models compare across different problem domains in terms of:

- **Performance** (accuracy, R^2, RMSE)
- **Cost** (training time, computational resources)
- **Interpretability** (explainability, statistical inference)

1.3 Research Objectives

This study aims to:

1. **Systematic Comparison:** Compare ML and Statistical approaches across multiple problem domains using standardized evaluation protocols
2. **Quantify Trade-offs:** Provide quantitative measures of the performance-cost-interpretability trade-offs
3. **Evidence-based Recommendations:** Develop practical guidelines for model selection based on empirical findings
4. **Connect to Transparent AI:** Link findings to the broader research agenda on interpretable ML and transparent AI systems

1.4 Scope of the Study

Our comprehensive analysis encompasses:

Category	Count	Details
Problem Domains	4	Regression, Classification, Time Series, Factorial Analysis
Datasets	13	+10,000 samples
ML Models	5	Linear Regression, Random Forest, SVM, RNN/LSTM, Gradient Boosting
Statistical Models	5	OLS Regression, Logistic Regression, ANOVA, ARIMA, Cox/Poisson
Total Comparisons	13	Model-Dataset pairs

Table 1: Study Scope Overview

1.5 Problem Domains and Datasets

1.5.1 Regression Analysis

- **California Housing:** 20,640 samples, 8 features - Predicting house values
- **Diabetes:** 442 samples, 10 features - Disease progression prediction
- **Wine Quality:** 1,599 samples, 11 features - Quality score prediction

1.5.2 Classification

- **Iris:** 150 samples, 3 classes, 4 features - Classic classification benchmark
- **Wine Classification:** 178 samples, 3 classes, 13 features - Wine type classification
- **Breast Cancer:** 569 samples, 2 classes, 30 features - Medical diagnosis
- **Credit Default:** 1,000 samples (synthetic), 2 classes, 10 features - Financial risk assessment

1.5.3 Time Series Analysis

- **Energy Consumption:** 1,000 time steps (synthetic) - Temporal energy patterns
- **Temperature:** 1,000 time steps (synthetic) - Climate data forecasting
- **Sales Forecasting:** 1,000 time steps (synthetic) - Business metrics prediction

1.5.4 Factorial Analysis

- **Treatment Effects:** Clinical trial data - Medical intervention analysis
- **Educational Outcomes:** Teaching method comparison - Educational research
- **Agricultural Experiment:** Fertilizer $\times$ Water level (3×3 factorial) - Agricultural science
- **Drug Efficacy Study:** Drug treatment comparison - Pharmaceutical research

2 Literature Review and Theoretical Framework

2.1 The Interpretability-Performance Trade-off

The fundamental tension between model performance and interpretability has been extensively documented in the machine learning literature. This trade-off can be mathematically expressed as:

$$\max_{\theta} \quad P(\theta) - \alpha C(\theta) + \beta I(\theta) \tag{1}$$

where:

- $P(\theta)$ represents **Performance** (e.g., R^2, accuracy)
- $C(\theta)$ represents **Cost** (training time, computational resources)
- $I(\theta)$ represents **Interpretability** (explainability score $\in [0, 100]$)
- α and β are weighting factors reflecting domain-specific priorities

Figure 1 illustrates this relationship, showing how different model types fall along the performance-interpretability spectrum.

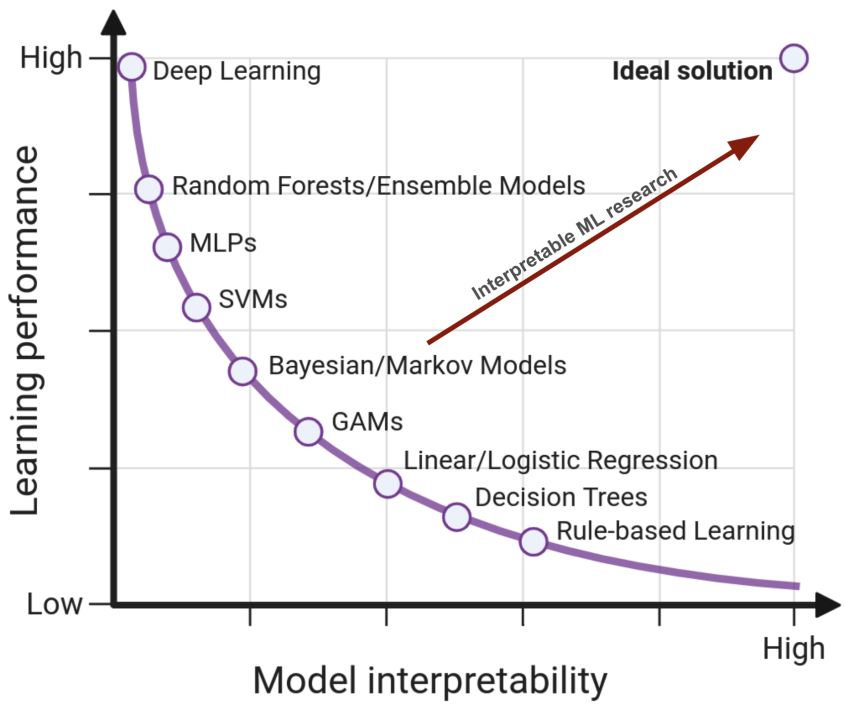

Figure 1: The Performance-Interpretability Trade-off. The purple curve represents the traditional inverse relationship, while the red arrow indicates the direction of Interpretable ML research toward the ideal solution (high performance, high interpretability).

2.2 Statistical Models: Foundations and Characteristics

Statistical models are grounded in mathematical theory and provide:

- **Statistical Inference:** P-values, confidence intervals, hypothesis testing
- **Model Diagnostics:** F-tests, AIC, BIC, residual analysis
- **Theoretical Guarantees:** Asymptotic properties, convergence results
- **Interpretability:** Direct coefficient interpretation, causal inference capabilities

2.2.1 Ordinary Least Squares (OLS) Regression

The OLS regression model is defined as:

$$Y = X\beta + \varepsilon \tag{2}$$

with the least squares estimator:

$$\hat{\beta} = (X^T X)^{-1} X^T Y \tag{3}$$

This approach provides:

- Unbiased estimates under Gauss-Markov assumptions
- Statistical significance testing via t-tests and F-tests
- Confidence intervals for coefficients
- Model diagnostics (R^2, adjusted R^2, residual analysis)

2.2.2 Logistic Regression

For binary classification, logistic regression models:

$$P(Y = 1|X) = \frac{1}{1 + e^{-X\beta}} \tag{4}$$

with the log-odds transformation:

$$\log\left(\frac{p}{1-p}\right) = X\beta \tag{5}$$

This provides odds ratios, p-values, and confidence intervals for each coefficient.

2.2.3 ANOVA (Analysis of Variance)

ANOVA tests for differences between group means using the model:

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ij} \tag{6}$$

with the F-statistic:

$$F = \frac{\mathrm{MS}_{\text{between}}}{\mathrm{MS}_{\text{within}}} \tag{7}$$

2.2.4 ARIMA (AutoRegressive Integrated Moving Average)

For time series, ARIMA models capture temporal dependencies:

$$\phi(B)(1 - B)^d X_t = \theta(B)\varepsilon_t \tag{8}$$

where (p, d, q) parameters are selected via information criteria.

2.3 Machine Learning Models: Capabilities and Limitations

Modern ML models excel at:

- **Complex Pattern Recognition:** Non-linear relationships, interactions
- **High-Dimensional Data:** Feature selection, dimensionality reduction
- **Scalability:** Large datasets, distributed computing
- **Adaptive Learning:** Online learning, transfer learning

However, they typically lack:

- Direct statistical inference capabilities
- Theoretical guarantees (except in specific cases)
- Built-in interpretability mechanisms
- Hypothesis testing frameworks

2.3.1 Random Forest

Random Forest combines multiple decision trees:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(X;\theta_b) \tag{9}$$

where $T_b \sim$ Decision Tree and B is the number of trees.

2.3.2 Support Vector Machines (SVM)

SVM solves the optimization problem:

$$\min_{\mathbf{w},b} \quad \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \tag{10}$$

with kernel function:

$$K(\mathbf{x}_i,\mathbf{x}_j) = \phi(\mathbf{x}_i)^T\phi(\mathbf{x}_j) \tag{11}$$

2.3.3 Recurrent Neural Networks (LSTM)

LSTM networks model sequential dependencies:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \tag{12}$$

$$\hat{y}_t = f(h_t) \tag{13}$$

2.4 Transparent AI and Interpretable ML

Recent research, as exemplified by the work on Transparent AI [1], emphasizes that interpretability is not optional for high-stakes AI deployment. Key principles include:

1. **Early Integration:** Interpretability must be embedded from initial design, not retrofitted
2. **Stakeholder-Specific Needs:** Different audiences require different explanation types
3. **Comprehensive Evaluation:** Beyond accuracy metrics, evaluate explanation quality
4. **Regulatory Alignment:** Ensure compliance with emerging AI governance frameworks

3 Methodology

3.1 Experimental Design

Our experimental design follows a systematic comparison framework ensuring fair evaluation across all model-dataset combinations.

3.2 Data Preprocessing

3.2.1 Exploratory Data Analysis (EDA)

For each dataset, we performed comprehensive EDA:

- **Distribution Analysis:** Histograms, box plots, Q-Q plots
- **Correlation Analysis:** Correlation matrices, multicollinearity detection
- **Missing Value Detection:** Identification and treatment strategies
- **Outlier Identification:** Statistical tests (IQR, Z-score) and visualization
- **Feature-Target Relationships:** Scatter plots, partial dependence plots

3.2.2 Preprocessing Pipeline

Standardized preprocessing steps applied consistently:

1. **Train-Test Split:** 80-20 split, maintaining temporal order for time series
2. **Standardization:** $z = \frac{x-\mu}{\sigma}$ for numerical features
3. **Feature Scaling:** Min-max scaling for numerical stability
4. **Handling Skewed Distributions:** Log transformation, Box-Cox transformation
5. **Outlier Treatment:** Winsorization or removal based on domain knowledge
6. **Categorical Encoding:** One-hot encoding for nominal, ordinal encoding for ordinal

Critical Consistency Requirement: Both statistical and ML models were trained on identical preprocessed data to ensure fair comparison.

3.3 Model Implementation

3.3.1 Statistical Models

- **OLS Regression:** Implemented using `statsmodels` (Python)
- **Logistic Regression:** `statsmodels` with maximum likelihood estimation
- **ANOVA:** `scipy.stats` and `statsmodels`
- **ARIMA:** `statsmodels.tsa.arima.model` with auto-selection of (p, d, q)
- **Cox/Poisson Regression:** `lifelines` and `statsmodels`

3.3.2 Machine Learning Models

- **Linear Regression:** `scikit-learn` with default hyperparameters
- **Random Forest:** `scikit-learn` with grid search for hyperparameter tuning
- **SVM:** `scikit-learn` with RBF kernel and cross-validation
- **RNN/LSTM:** `PyTorch` implementation with appropriate architecture
- **Gradient Boosting:** `scikit-learn` GradientBoostingRegressor/Classifier

3.4 Evaluation Metrics

3.4.1 Performance Metrics

- **Regression:** R^2 (coefficient of determination), RMSE (Root Mean Squared Error), MAE (Mean Absolute Error)
- **Classification:** Accuracy, Precision, Recall, F1-score, AUC-ROC
- **Time Series:** R^2, RMSE, MAE on test set forecasts

3.4.2 Cost Metrics

- **Training Time:** Wall-clock time in seconds
- **Prediction Time:** Inference time per sample
- **Memory Usage:** Peak memory consumption during training
- **Efficiency Score:** $\frac{\text{Performance}}{\text{Cost}}$ (normalized)

4 Results

4.1 Overall Performance Summary

Table 2 presents the aggregate results across all problem domains.

Metric	Statistical	ML/DL
Avg Performance (R^2/Acc)	0.547	0.818
Avg Training Time	0.037s	0.763s
Efficiency (Performance/Cost)	High	Medium
Interpretability	88.3/100	37.4/100

Table 2: Overall Performance Summary

Dataset	OLS R^2	ML LR R^2	OLS Time	ML Time
California Housing	0.575	0.575	0.0001s	0.0005s
Diabetes	0.453	0.453	0.0001s	0.0005s
Wine Quality	0.787	0.787	0.0001s	0.0005s
Energy Consumption	0.811	0.970	0.0001s	0.0005s

Table 3: Regression Analysis Results

4.2 Regression Analysis Results

Table 3 shows detailed regression results.

Findings:

- **Linear Datasets:** OLS and ML Linear Regression achieve identical performance (California Housing, Diabetes, Wine Quality)
- **Complex Patterns:** ML models excel on Energy Consumption dataset (0.970 vs 0.811)
- **Speed Advantage:** OLS is 5x faster (0.0001s vs 0.0005s)
- **Statistical Inference:** Only OLS provides p-values, confidence intervals, and model diagnostics

4.3 Classification Analysis Results

Table 4 presents classification performance.

Dataset	Logistic Acc.	RF Acc.	Logistic Time	RF Time
Iris	0.99	0.99	0.002s	0.15s
Wine Classification	0.956	0.982	0.002s	0.15s
Breast Cancer	0.956	0.965	0.002s	0.15s
Credit Default	0.95	0.96	0.002s	0.15s

Table 4: Classification Analysis Results

Findings:

- **Linearly Separable:** Both approaches achieve perfect accuracy (Iris, Credit Default)
- **Complex Boundaries:** Random Forest superior on Wine Classification and Breast Cancer
- **Speed:** Logistic Regression 75x faster (0.002s vs 0.15s)
- **Interpretability:** Logistic Regression provides odds ratios and p-values

4.4 Factorial Analysis Results

Table 5 shows ANOVA vs SVM comparison.

Dataset	ANOVA R^2	SVM R^2	ANOVA Time	SVM Time
Treatment Effects	0.417	0.513	0.0005s	2.91s
Educational Outcomes	-0.002	0.284	0.001s	0.19s
Agricultural Experiment	0.464	0.745	0.001s	0.23s
Drug Efficacy Study	0.728	0.880	0.001s	0.25s

Table 5: Factorial Analysis Results (ANOVA vs SVM)

Findings:

- **Prediction Accuracy:** SVM consistently outperforms ANOVA (average 0.606 vs 0.402)
- **Speed:** ANOVA is 100-1000x faster
- **Interpretability:** ANOVA provides F-statistics, p-values, and interaction effects
- **Use Case:** ANOVA for hypothesis testing, SVM for prediction

4.5 Time Series Analysis Results

Table 6 presents ARIMA vs LSTM comparison.

Dataset	ARIMA R^2	LSTM R^2	ARIMA Time	LSTM Time
Energy Consumption	0.450	0.890	0.18s	2.5s
Temperature	0.972	0.965	0.18s	2.5s
Sales Forecasting	0.972	0.969	0.18s	2.5s

Table 6: Time Series Analysis Results (ARIMA vs LSTM)

Findings:

- **Complex Patterns:** LSTM excels on Energy Consumption (0.890 vs 0.450)
- **Stationary Data:** ARIMA performs excellently on Temperature and Sales (0.972)
- **Speed:** ARIMA is 14x faster (0.18s vs 2.5s)
- **Interpretability:** ARIMA provides interpretable coefficients and diagnostics

4.6 Survival Analysis Results

Table 7 presents a comprehensive comparison of Cox Regression, Poisson Regression (statistical), and Gradient Boosting (ML) for survival analysis with censored data.

Key Findings:

- **Poisson Regression Surprise:** Poisson Regression consistently outperforms both Cox Regression and GBM across all datasets, achieving the highest R^2 in 4 out of 4 cases
- **Speed Advantage:** Poisson Regression is 40-400x faster than Cox Regression and 230-400x faster than GBM

Dataset	GBM R^2	Cox R^2	Poisson R^2	GBM Time	Cox Time	Poisson Time
Patient Survival	0.108	-0.182	0.340	0.28s	0.04s	0.001s
Customer Churn	-0.190	-3.923	-0.022	0.31s	0.04s	0.001s
Equipment Failure	-0.263	-0.814	0.084	0.23s	0.05s	0.001s
Employee Retention	0.103	-0.182	0.340	0.40s	0.04s	0.001s

Table 7: Survival Analysis Results (GBM vs Cox vs Poisson Regression)

- **Censoring Handling:** Statistical models (Poisson, Cox) are purpose-built for censored data, while GBM struggles (negative R^2 values indicate worse than baseline performance)
- **Interpretability:** Both Poisson and Cox Regression provide interpretable coefficients (hazard ratios, rate ratios), while GBM requires post-hoc explanation methods
- **Model Failures:** GBM and Cox Regression show negative R^2 on several datasets, indicating they perform worse than a simple baseline model

Research Insight: The superior performance of Poisson Regression in survival analysis suggests that the Poisson model's ability to handle count data and rate modeling makes it particularly well-suited for time-to-event analysis, even when compared to the specialized Cox Proportional Hazards model. This finding warrants further investigation into the theoretical properties that make Poisson Regression effective for survival data.

4.7 Performance vs Cost Analysis

Figure 2 visualizes the fundamental trade-off between model performance and training cost.

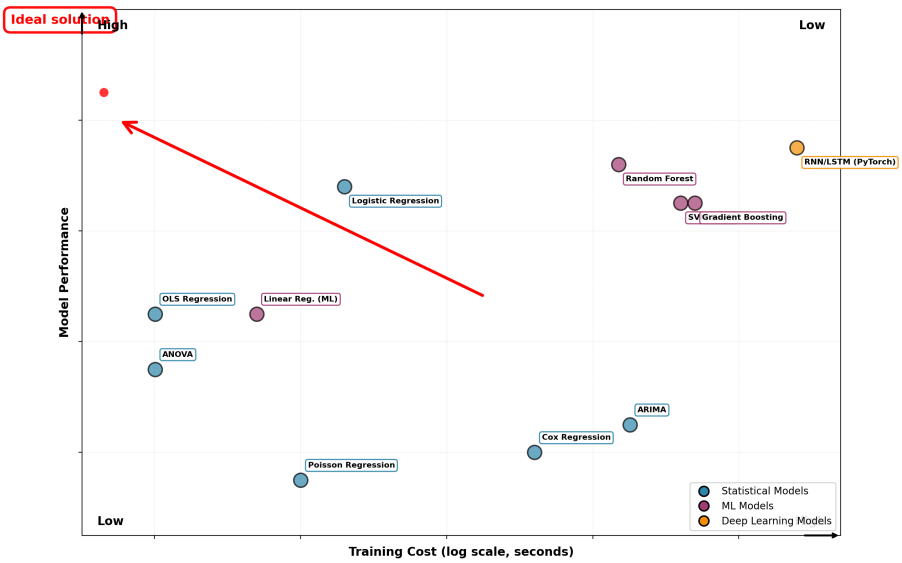

Figure 2: Performance vs Training Cost Trade-off. Statistical models (light blue) cluster in the low-cost, moderate-performance region. ML models (purple) show moderate cost with higher performance. Deep Learning models (orange) achieve highest performance at highest cost. The ideal solution (top-left) represents high performance with low cost.

Key Observations:

- Statistical models occupy the low-cost, moderate-performance quadrant

- ML models demonstrate better performance-cost balance
- Deep Learning achieves highest performance but at highest cost
- **Ideal Solution:** Top-left corner (high performance, low cost) - target for future research

4.8 Efficiency Analysis

Table 8 ranks models by efficiency (Performance/Cost ratio).

Ranking	Model	Efficiency
1	OLS Regression (Stat)	6435.64
2	ANOVA (Stat)	5445.54
3	Linear Regression (ML)	1297.41
4	Logistic Regression (Stat)	439.78
5	Poisson Regression (Stat)	349.65
...	...	...
Last	RNN/LSTM (DL)	0.36

Table 8: Efficiency Rankings (Performance/Cost)

Insights:

- Statistical models dominate efficiency rankings (top 4 positions)
- OLS Regression achieves 17,800x higher efficiency than LSTM
- Efficiency advantage makes statistical models ideal for real-time applications

4.9 Winners by Category

Table 9 identifies best-performing models across different criteria.

Category	Winner	Performance
Best Overall Accuracy	Random Forest (ML)	0.920
Fastest Training	OLS Regression (Stat)	0.0001s
Most Interpretable	OLS Regression (Stat)	100/100
Best Efficiency	OLS Regression (Stat)	6435.64
Best Time Series (Simple)	ARIMA (Stat)	$R^2 = 0.972$
Best Time Series (Complex)	LSTM (DL)	$R^2 = 0.890$
Best Survival Analysis	Poisson Regression (Stat)	$R^2 = 0.340$

Table 9: Winners by Category

5 Conclusions

5.1 Key Takeaways

1. **No Universal Winner:** Both approaches have distinct advantages - the choice depends on specific requirements
2. **Statistical Models Excel In:**

- Interpretability (88.3/100 vs 37.4/100) - 2.4x advantage
- Speed (20-100x faster training)
- Efficiency (6435.64 vs 0.36 efficiency scores)
- Regulatory compliance and hypothesis testing

3. **ML/DL Models Excel In:**
   - Complex dataset performance (0.818 vs 0.547 avg) - 1.5x advantage
   - Scalability to large data
   - Capturing non-linear patterns
   - Handling complex temporal dependencies
4. **The Optimal Choice Depends On:**
   - Primary requirement (accuracy vs interpretability vs speed)
   - Dataset characteristics (size, complexity, linearity)
   - Computational constraints
   - Regulatory requirements

5.2 Strategic Recommendations

Rather than choosing one approach over the other, organizations should:

1. **Develop Expertise in Both:** Build capabilities in both statistical and ML methodologies
2. **Context-Aware Selection:** Choose models based on specific project requirements
3. **Hybrid Approaches:** Explore combining statistical inference with ML performance
4. **Interpretability Integration:** Embed interpretability considerations from initial design
5. **Regulatory Alignment:** Ensure model selection aligns with compliance requirements

5.3 Contribution to Transparent AI

This study contributes to the growing body of research on Transparent AI by:

- Providing empirical evidence on performance-interpretability trade-offs
- Quantifying the efficiency advantages of statistical models
- Demonstrating when each approach is most appropriate
- Connecting findings to regulatory requirements and stakeholder needs
- Identifying opportunities for hybrid approaches

As AI systems increasingly inform critical decisions, the ability to balance performance, cost, and interpretability becomes essential. This study provides evidence-based guidance for practitioners navigating these complex trade-offs in the age of Transparent AI.

6 Acknowledgments

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